Problem Set #6 - Answer Sheet

Written problems

1. $LL(\lambda) = n \log \lambda - \sum_{i=1}^{n} \lambda x_i$

$$L(\lambda) = \prod_{i=1}^{n} f(X_i = x_i \mid \Lambda = \lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i}$$

$$LL(\lambda) = \sum_{i=1}^{n} \log \lambda e^{-\lambda x_i} = \sum_{i=1}^{n} (\log \lambda + (-\lambda x_i)) = n \log \lambda - \sum_{i=1}^{n} \lambda x_i$$

2.

The Naive Bayes is based on MAP, but makes the assumption that all features of X are independent.

According to MAP:

$$\hat{y} = \arg \max_{y \in \{0,1\}} P(Y = y \mid X = x) = \arg \max_{y \in \{0,1\}} \frac{P(X = x \mid Y = y)P(Y = y)}{P(X = x)} = \arg \max_{y \in \{0,1\}} P(X = x \mid Y = y)P(Y = y)$$

Based on the assumption the Naive Mayes

$$\hat{y} = \arg \max_{y \in \{0,1\}} P(X = x \mid Y = y)P(Y = y) = \arg \max_{y \in \{0,1\}} \prod_{i=1}^{n} P(X_i = x_i \mid Y = y)P(Y = y) = \arg \max_{y \in \{0,1\}} \sum_{i=1}^{n} (\log P(X_i = x_i \mid Y = y) + \log P(Y = y))$$

If the first k data are all the same, then the probability of values for the first k variable will be the same as the first variable. Not their actual probability. Because the Naive Bayes Classifier uses the latter (their actual probability) to classify, then the result will be faulty.

Programming problems

(See the code file)